

E-Commerce Data Analysis: Uncovering Sales Trends & Business Insights

Project Overview

Objective

Analyze valid e-commerce transactions to identify sales trends and category performance across 2021–2022.

Dataset

Tokopedia simulated dataset 4 relational tables

Tools



PostgreSQL



Python



Looker Studio



Step 01: Discover
SQL Analysis

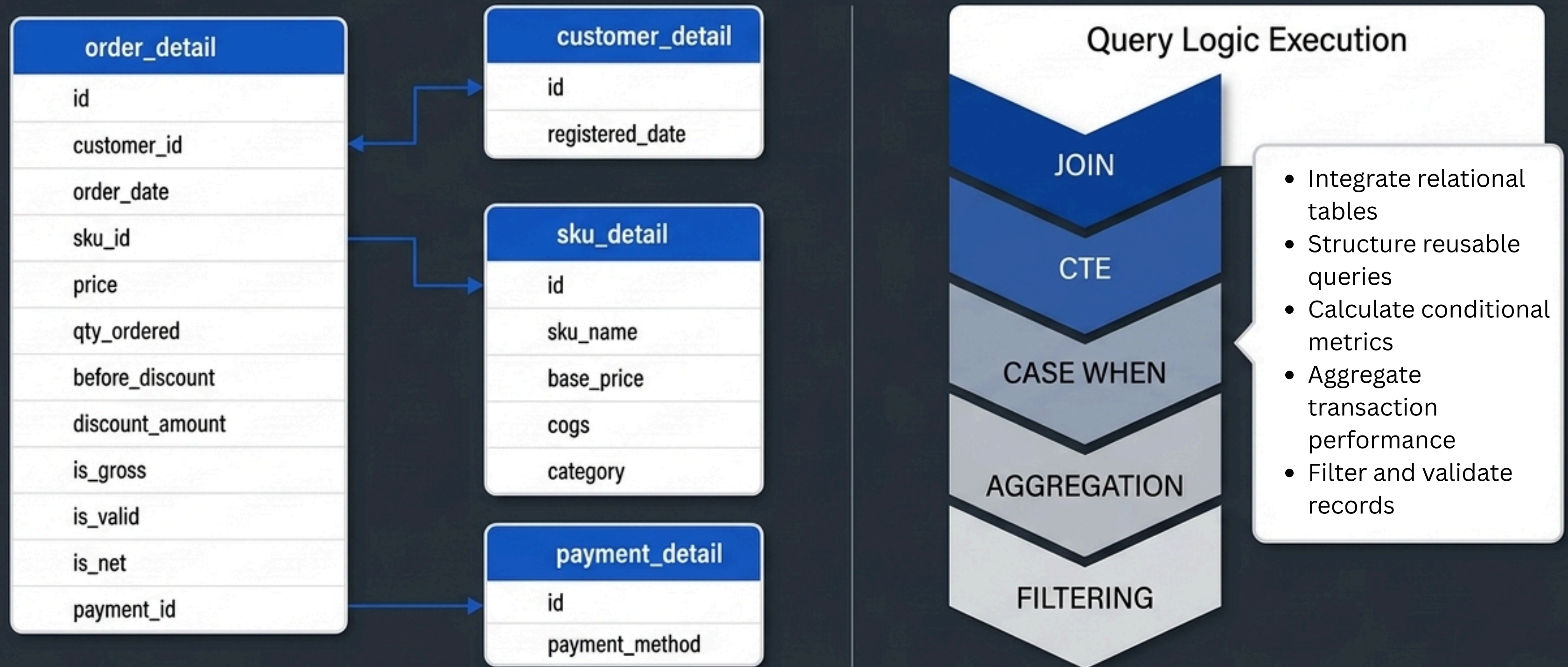


Step 02: Investigate
Python Analysis



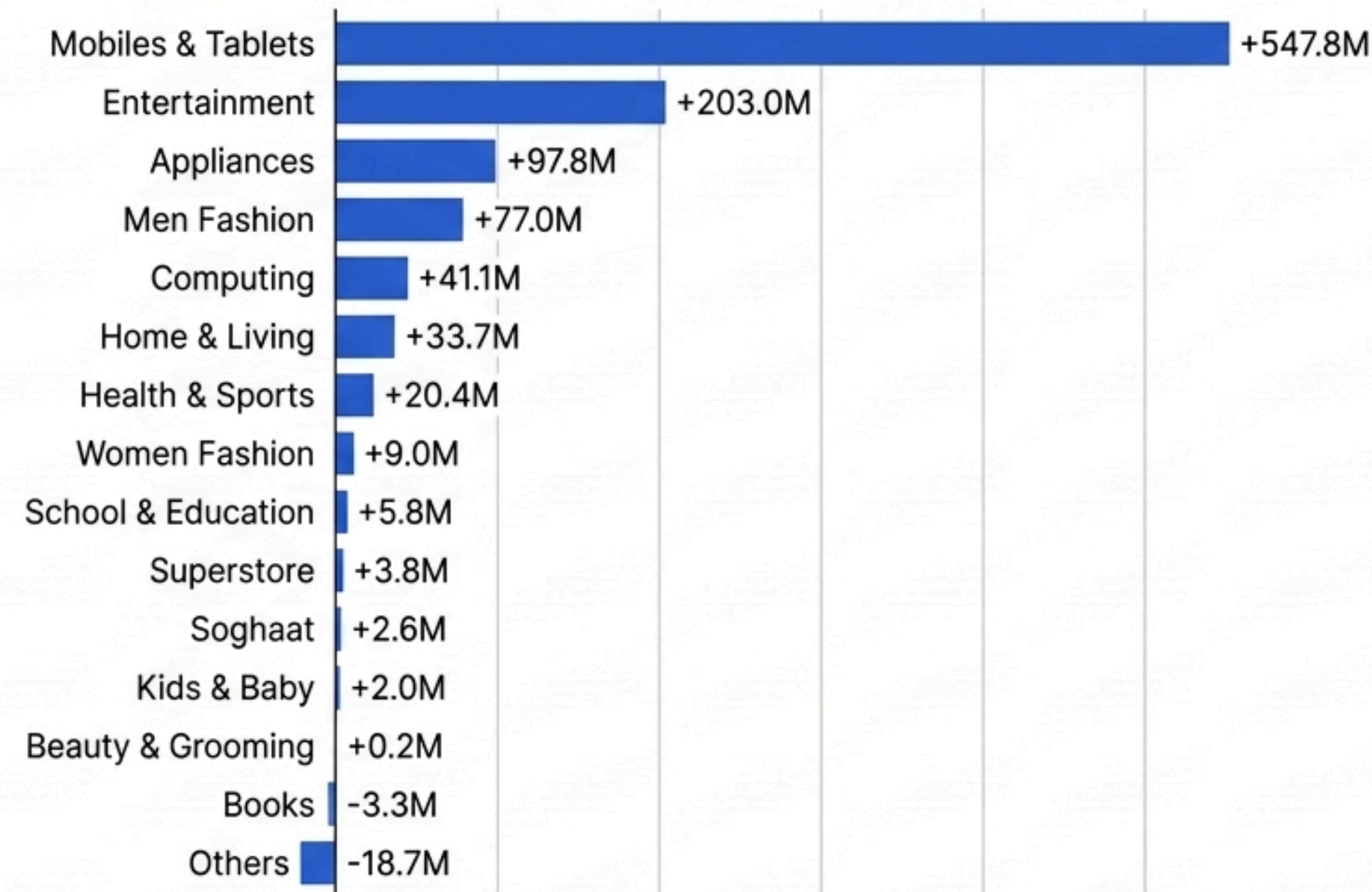
Step 03: Communicate
Business Insights

Translating Raw E-Commerce Data into Structured Relational Models



High-Level Category Performance and Transaction Growth (2021–2022)

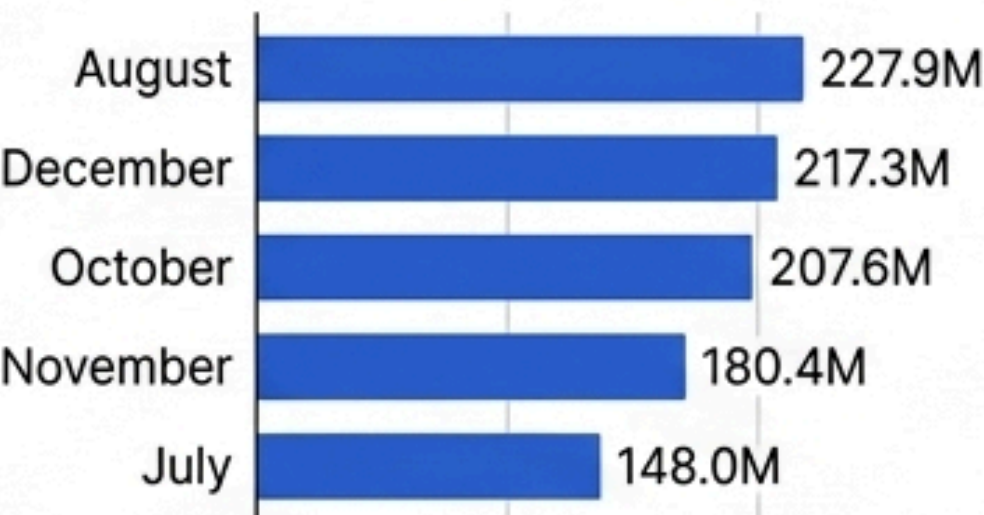
Transaction Value Growth by Category — 2021 to 2022



13 Up / 
2 Down 

Categories exhibiting positive vs. negative transaction growth.

Top 5 Monthly Transaction Values — 2021



Diagnostic Deep Dive: Investigating the 'Others' Category Decline

The Evidence

 pandas,  NumPy, matplotlib, seaborn



The Problem & Takeaway

-163 units
(~38.3% Decline)

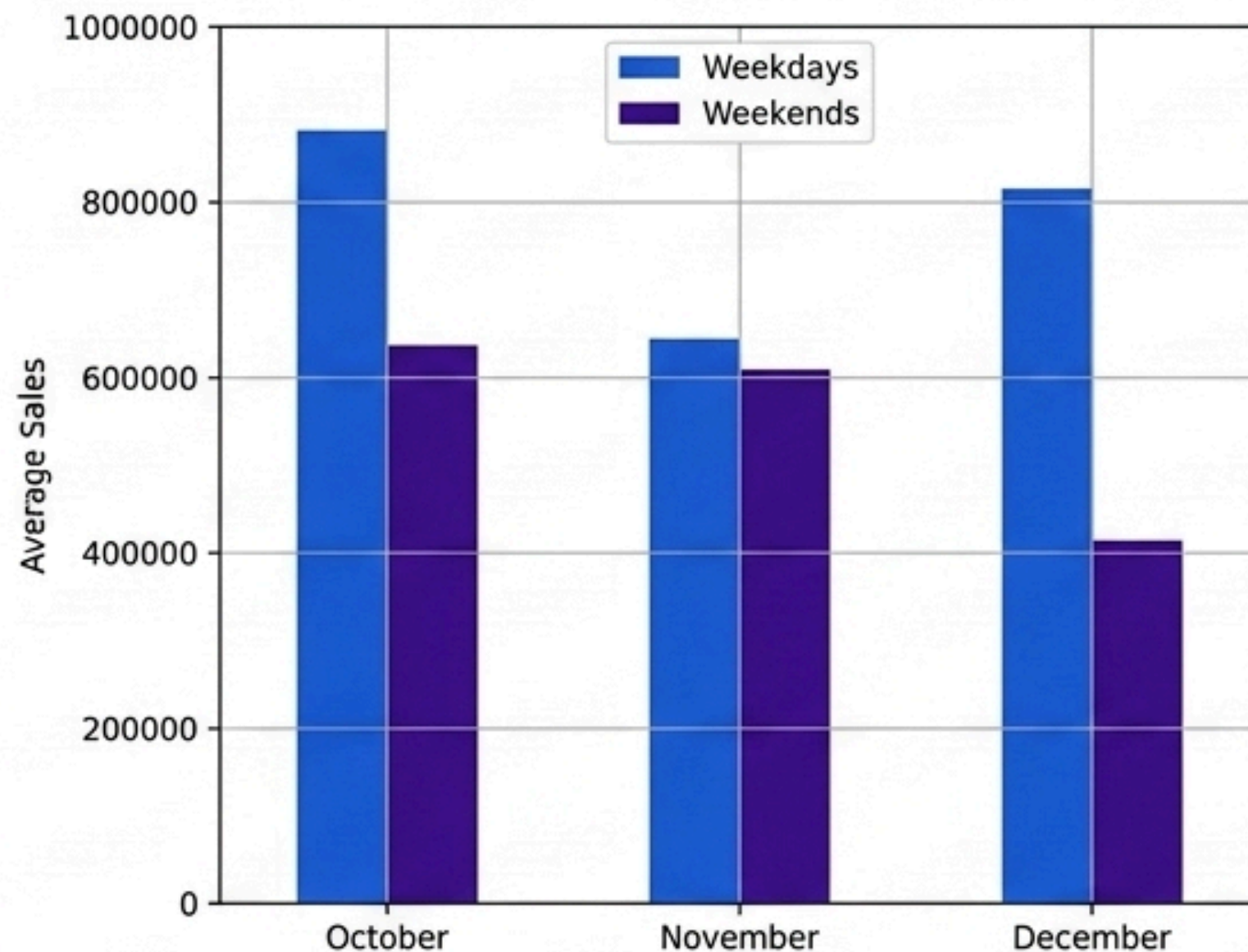
Sales quantity collapse from 2021 to 2022.

Business Takeaway

Identify declining categories and investigate their underlying sales performance before planning inventory actions.

Evaluating Campaign Effectiveness: Did Weekends Actually Drive Sales?

Weekends vs Weekdays Average Daily Sales: Oct-Dec



\$211,280K

Average Sales Gap

Weekday average sales were significantly higher than weekend sales during the Q4 2022 campaign period.

Business Takeaway

Reassess weekend campaign timing and strategy based on actual sales response data.

Executive Dashboard & Macro Business Insights

Campaign Performance Dashboard

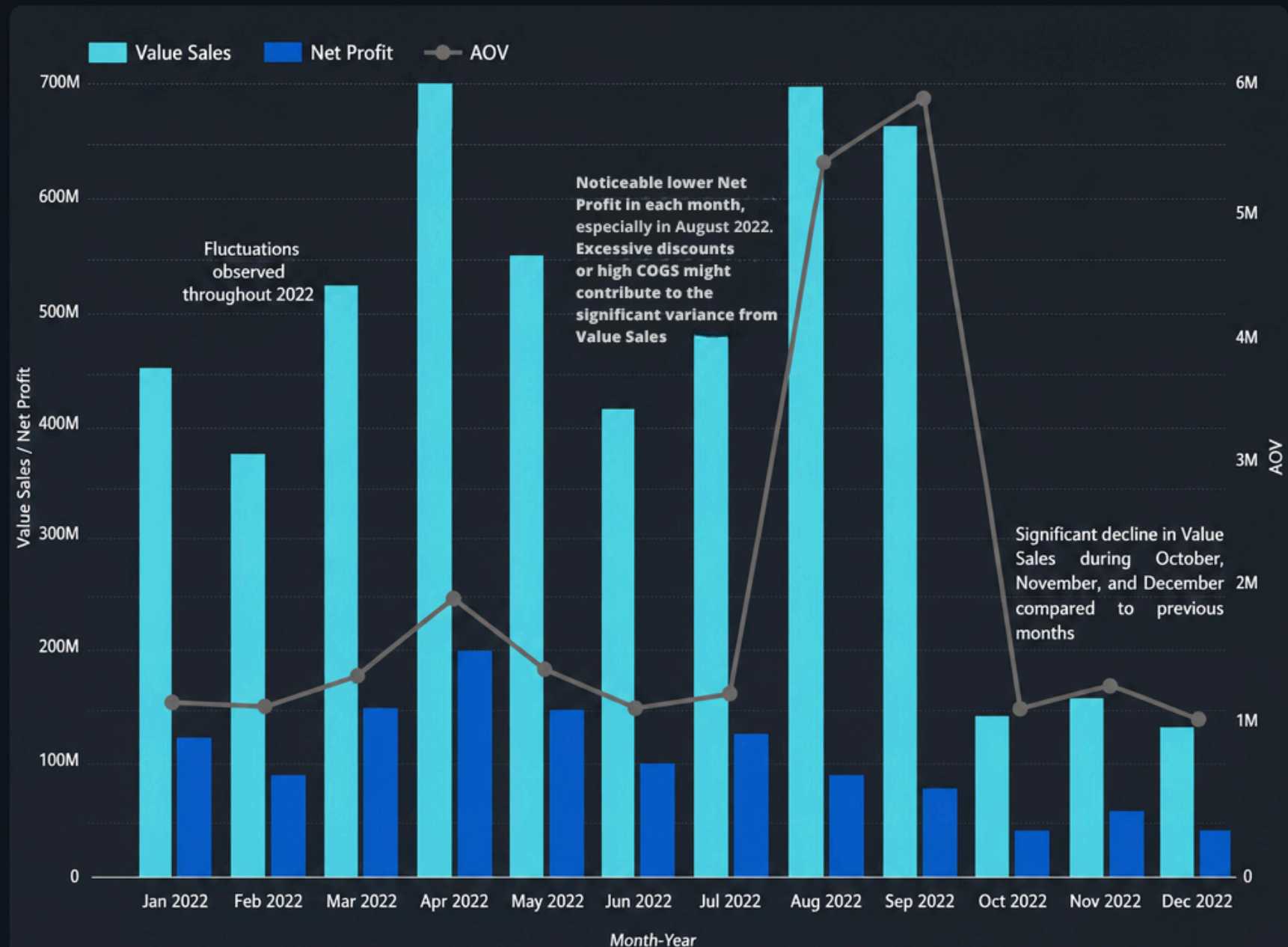
Jan 1, 2022 - Dec 31, 2022

Category

Payment Method

Value Transaction

Relationship Analysis: Value Sales, Net Profit, and AOV Trends (2022)



Fluctuating Performance

Value sales and net profit fluctuated heavily throughout 2022.

Sales-Profit Gap

A persistent gap between sales and profit indicates severe pressure from COGS and discount strategies.

Year-End Decline

Value sales declined sharply in Q4 (October to December 2022).

The Golden Rule

High Sales ≠ High Profit.

Revenue growth does not always translate into bottom-line profit.

Strategic Recommendations & Project Assets

Review COGS

Identify specific cost components crushing profit margins.

Optimize Discounts

Evaluate the true ROI of promotional strategies and discount utilization.

Investigate the Decline

Explore the underlying systemic factors behind the October–December sales drop.

Reassess Campaigns

Pivot Q4 campaign timing away from weekends based on actual A/B sales responses.

Additional Targeting & Python Findings

Top 5 products in Mobile & Tablets flagged for Year-End Festival prizes.

Retargeting protocols established for 2022 abandoned checkouts.

SQL

Data Extraction & Transformation

Python

Analysis & Visualization

Looker Studio

Interactive Dashboard

[Full Project Documentation](#)